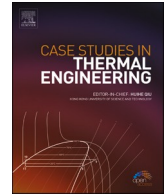




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## Thermal management inside a discretely heated rectangular cuboid using P, PI and PID controllers

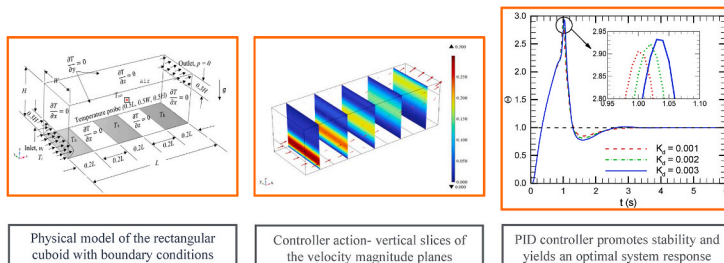
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## GRAPHICAL ABSTRACT

## Thermal Management Inside a Discretely Heated Rectangular Cuboid Using P, PI and PID Controllers

System's response at the probe point using different combinations of proportional (P), integral (I), and derivative (D) controllers to analyze and compare the system's steady-state error and transient behavior



## ARTICLE INFO

## Keywords:

Cuboid  
Steady-state error  
Discrete heating  
PID controller  
Thermal management system

## ABSTRACT

This study investigates the viability of three different continuous (P, PI, and PID) controllers to meet specific thermal requirements at a desired location in a cooling system with discrete heat sources. The system is a rectangular cuboid with three discrete heat sources placed on the bottom surface at periodic intervals, and the rest of the walls are insulated. A temperature probe is installed in the system's center to monitor the temperature and provide feedback to the flow controllers in a continuous manner. The velocity of air entering from the inlet port varies in

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response to the controller feedback and discharge through the outlet port at atmospheric condition. The Galerkin finite element technique solves the governing Navier-Stokes and energy equations and the appropriate initial and boundary conditions. The simulations consist of testing the system's response at the probe point using different combinations of proportional (P), integral (I), and derivative (D) controllers with varied gains to analyze and compare the system's steady-state error and transient behavior in terms of overshoot, oscillation, and settling time. The results indicate that the P controller cannot eliminate steady-state error, while the PI controller achieves zero steady-state error and faster settling. However, if appropriately tuned, the PID controller enhances oscillation control and response time.

## 1. Introduction

The use of external power to enhance thermal performance has been a critical approach in various applications, including power plants [1], food storage [2], and micro-electromechanical systems (MEMS) devices [3], among others. In electronic devices, where effective cooling is becoming increasingly important, mixed convection is being incorporated more frequently to improve performance [4–6]. Therefore, it is crucial to consider the impact of mixed convection when designing and analyzing cooling systems to optimize thermal performance. It requires a thorough understanding of the fluid flow and heat transfer characteristics and the use of numerical simulations, experimental measurements, and analytical models to accurately predict heat transfer rates in mixed convective flows [7, 8]. The absence of an appropriate temperature control system can result in various adverse outcomes such as equipment damage, energy wastage, compromised product quality, and fire hazards [9–11]. Temperature regulation can be accomplished by implementing various methods, including air conditioning, proper insulation, and ventilation systems. It can be achieved by utilizing temperature detection sensors like thermistors to monitor the environment and facilitate temperature control.

Proportional ( $K_p$ ), integral ( $K_i$ ), and derivative ( $K_d$ ) gains are essential components of the P, PI (proportional-integral), and PID (proportional-integral-differential) controllers, which are widely used in continuous feedback control systems [12,13]. Proportional gain generates a control signal proportional to the error. Integral gain reduces steady-state error by integrating accumulated error, whereas derivative gain predicts future error trends to prevent overshooting. Besides, the PI controller uses both proportional and integral gains to reduce steady-state error, and the PID controller utilizes all three gains for faster response, reduced overshooting, and minimized steady-state error. The selection and tuning of P, I, and D gains depend on the system's characteristics. The PID controller is utilized in industrial and research settings to enhance the precision of temperature regulation [14–16]. It has been found in many practical applications in various fields, such as maintaining the temperature at a set point in friction stir welding [17], regulating the water temperature in electric kettles [18], and adapting to weather and climate changes [19]. Grassi and Tsakalis [20] employed a frequency loop-shaping-based PID tuning approach to managing the temperature of diffusion/chemical vapor deposition furnaces utilized in semiconductor production. Durand-Estebe et al. [21] investigated the optimal cooling temperature that would achieve a balance between the energy consumption of the chiller and the server. Lee and Chen [22] found that a self-tuning PID controller could save 14% of a server's fan power consumption if a bit of overshoot of temperature response were permitted. Song et al. [23] proposed a novel positive temperature coefficient material combined with a PID controller for better thermal management of electronic devices. By selecting the appropriate gains and tuning the controller, a feedback control system can achieve precise and stable control over the process, which is crucial in various engineering applications [24–26].

The exploration of convection in vented cavities has been the subject of many research papers. Saeidi and Khodadadi [27] examined the thermal flow in a vented square cavity with incoming flow oscillation. The outlet port's positioning was also considered in a study conducted by Sourtiji et al. [28], which investigated the heat transfer of  $\text{Al}_2\text{O}_3$ -water nanofluid in a ventilated cavity. Campo and Sieries [29] investigated the effect of wall temperature on the triangular cavity representing the sealed attic space in a house for different weather conditions, and their results were validated against the experimental results. The use of P, PI, and PID controllers in regulating the input air velocity in a T-shaped vented cavity was studied by Saha et al. [30]. After that, Durgam et al. [31] explored the optimum position of a heat source array on a substrate board to minimize the maximum excess temperature and found that a geometric distance parameter of  $\lambda = 1.3$  provided the best heat dissipation configuration. Shivran et al. [32] conducted a numerical analysis to investigate how different positions of inlet and outlet ports affected heat transfer in a square cavity filled with Cu-water nanofluid. Recently, Shuvo et al. [33] explored the effect of a hot cylinder in a vented enclosure to study the temperature response using continuous closed-loop (P, PI, and PID) controllers.

A more realistic perspective can be attained through analysis employing three-dimensional (3D) geometry [34,35]. Few studies have examined thermal performance in a vented 3D geometry. Jiang et al. [36] developed a new algorithm for temperature control utilizing fuzzy self-tuning PID, which had the potential to enhance system performance. Selimefendigil [37] studied a 3D vented-cubic structure with corrugated bottom wall conditions and an inner rotating adiabatic cylinder to examine its cooling performance. They found that the heat transfer rate only increased when the rotating cylinder spun anti-clockwise but enhanced for all conditions when the rotating cylinder was hot. Kareem et al. [38] conducted a numerical study of water within a closed lid-driven 3D cavity with a partially heated wall to investigate heat transfer by changing different parameters. Zisan et al. [39] used numerical methods to explore the natural convection within a cubic cavity with two heat sources on the bottom wall.

To the best of the authors' knowledge and based on the reviewed literature, no research has been conducted to investigate the impact of temperature response by regulating airflow at the inlet on a rectangular cuboid where the bottom wall is discretely heated. P, PI, and PID controllers are commonly employed to achieve accurate temperature control and precise airflow regulation through the

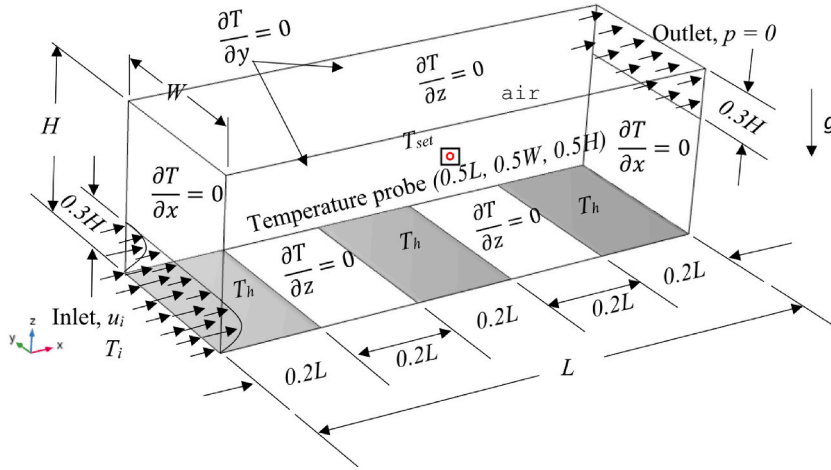


Fig. 1. Schematic sketch of the rectangular cuboid with appropriate dimensions and boundary conditions.

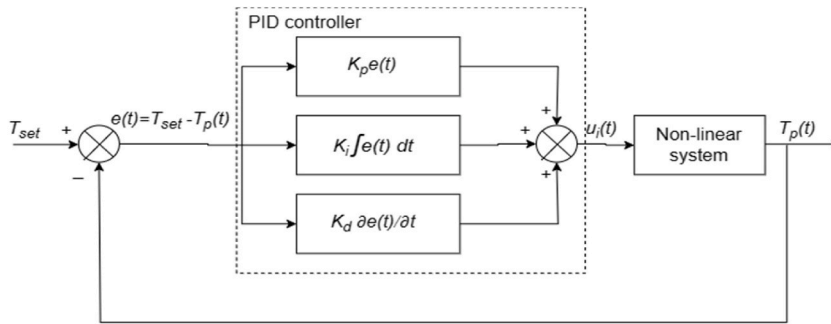


Fig. 2. Functional block diagram of the PID controller incorporated in this study.

inlet port. It ensures that the process operates within the desired range and remains stable. Other controllers, e.g., ID controllers, are limited in their inability to provide proportional action, resulting in potential sluggishness in responding to steady-state errors. Moreover, they carry the risk of integral windup when not tuned correctly, leading to overshoot or instability. Conversely, the PD controllers lack integral action, making them less effective in eliminating constant disturbances or steady-state errors. They are also sensitive to noise measurement, amplifying high-frequency variations, which can result in erratic control outputs. These limitations highlight why P, PI, and PID controllers are more versatile and commonly employed in various control scenarios. This phenomenon has various engineering applications, including developing efficient heat exchangers and thermal management systems. This research investigates the influence of different combinations of proportional, differential, and integral gains on a thermal management system's performance to maintain the required temperature at a desired location.

## 2. Physical modeling

The detailed geometry with necessary boundary conditions of the proposed computational domain (rectangular cuboid) is shown in Fig. 1. The rectangular cuboid's length, width, and height are  $L = 0.03$  m,  $W = 0.01$  m, and  $H = 0.01$  m, respectively. Three discrete, rectangular heating elements (strips) of equal size ( $0.2L \times W$ ) are placed at a regular interval of  $0.2L$  along its length in the bottom wall and maintained at a constant high temperature of  $T_h = 350$  K. The remaining walls are kept insulated. The domain is filled with air and has an initial temperature of  $T_0 = 290$  K. Air is selected as it is still one of the most preferred cooling media for electronic equipment like microcomputers due to its abundance and low cost [40–42]. The inside air content, flow condition, and temperature all change with time when the domain interacts through its inlet and outlet ports (equal size, rectangular ports of  $W \times 0.3H$ ) which are placed at the bottom and the top positions of the  $yz$  face of the rectangular cuboid as shown in Fig. 1. To maintain a constant temperature  $T_{set} = 300$  K at a particular set point ( $0.5L, 0.5W, 0.5H$ ), a temperature probe is inserted at the same point to monitor the time-varying probe-point temperature  $T_p(0.5L, 0.5W, 0.5H, t)$  and give feedback (error signal) to the controller. A controller action is activated based on this feedback ( $e(t)$ ) and adjusts the inflow conditions accordingly. The inflow condition of air is assumed to vary only in terms of flow velocity  $u_i = f(T_p)$  while keeping a fixed inlet temperature of  $T_i = 290$  K. The outflow condition is adjusted accordingly, ensuring a uniform, unidirectional flow at zero pressure. Moreover, the gravitational pull ( $g$ ) is assumed to act in the negative  $z$ -direction.

**Table 1**

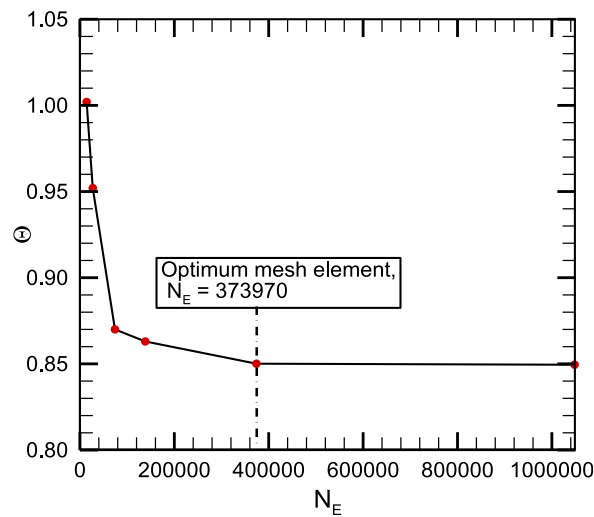
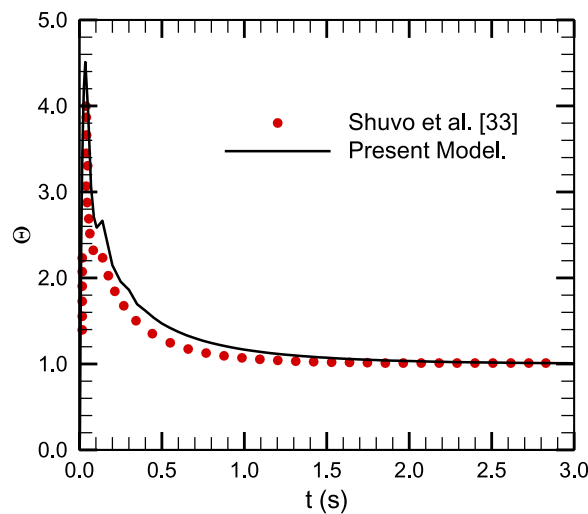
The thermo-physical properties of air at a mean temperature of 300 K [43].

| Properties         | $\rho$ [kgm <sup>-3</sup> ] | $k$ [Wm <sup>-1</sup> K <sup>-1</sup> ] | $c_p$ [Jkg <sup>-1</sup> K <sup>-1</sup> ] | $\mu$ [Pa.s]           | $\beta$ [K <sup>-1</sup> ] |
|--------------------|-----------------------------|-----------------------------------------|--------------------------------------------|------------------------|----------------------------|
| Values of interest | 1.184                       | 0.02551                                 | 1007                                       | $1.849 \times 10^{-5}$ | $3.33 \times 10^{-3}$      |

**Table 2**

Well-imposed initial and boundary conditions of the thermal and flow fields applied in the cuboid.

| Conditions          | Description     | Thermal field                                | Flow field                                                                                                                                                                                                      |
|---------------------|-----------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial ( $t = 0$ ) | Inside domain   | $T_0 = 290$ K                                | $u = v = w = 0$ ms <sup>-1</sup>                                                                                                                                                                                |
| Boundary            | Heater surface  | $T_h = 350$ K                                | $u = v = w = 0$ ms <sup>-1</sup>                                                                                                                                                                                |
|                     | Inlet           | $T_i = 290$ K                                | $u_i = K_p(T_p - T_{set}) + K_i(T_p - T_{set})dt + K_d\partial(T_p - T_{set})/\partial t$ ms <sup>-1</sup> if $T_p > T_{set}$ ; $u_i = 0$ ms <sup>-1</sup> if $T_p \leq T_{set}$ ; $v = w = 0$ ms <sup>-1</sup> |
|                     | Outlet          | $\partial T/\partial x = 0$ Km <sup>-1</sup> | $v = w = 0$ ms <sup>-1</sup> , $\partial u/\partial x = 0$ s <sup>-1</sup> , $p = 0$ Pa                                                                                                                         |
|                     | Remaining walls | $\partial T/\partial n = 0$ Km <sup>-1</sup> | $u = v = w = 0$ ms <sup>-1</sup>                                                                                                                                                                                |

**Fig. 3.** Mesh independence test in terms of the variation of non-dimensional temperature with respect to the number of mesh elements (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)**Fig. 4.** Verification of the present model by comparing the variation of  $\Theta$  with respect to  $t$  against the work of Shuvo et al. [33] (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

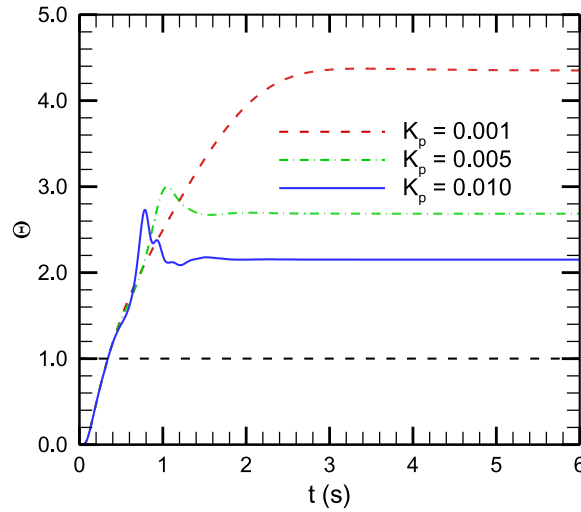


Fig. 5. System response in terms of  $\theta$  at the probe point with time due to P controller action for different  $K_p$  values (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In the current system, the controlled signal is obtained by summing the proportional, integral, and derivative of the error signal in the temporal domain. Therefore, it is referred to as a PID controller. Fig. 2 illustrates a typical functional block diagram of the PID controller employed in this study. In this model, both the measured and control variables exhibit a high degree of nonlinearity, posing significant challenges for control and stability analysis. However, when dealing with a nonlinear system, careful adjustments are necessary for implementing and tuning the gain terms ( $K_p$ ,  $K_i$ , and  $K_d$ ) to handle the nonlinear dynamics effectively. In more complex nonlinear problems, advanced control techniques such as model predictive control, adaptive control, and nonlinear control strategies are often more suitable.

In the work, the reasoning behind choosing specific gains for P, PI, and PID controllers is typically based on a trial-and-error approach that refines the initial approximations derived from the Ziegler-Nichols method, particularly in terms of their ratios. Suitable ratios are identified after numerous trials and errors, typically ranging from 1:50:1 to 3.333:66.667:1 for P, PI, and PID gains. Based on these ratios, the gains are then selected accordingly.

### 3. Mathematical modeling

The air is assumed to be a Newtonian fluid with constant thermo-physical properties with temperature. Boussinesq's approximation holds good to account for the volumetric body force induced due to density difference in terms of temperature difference. Neglecting the influence of internal heat generation and radiation, the governing equations for this time-dependent, laminar flow, mixed convection heat transfer problem are described in terms of the following three-dimensional continuity, momentum, and energy equations:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad (1)$$

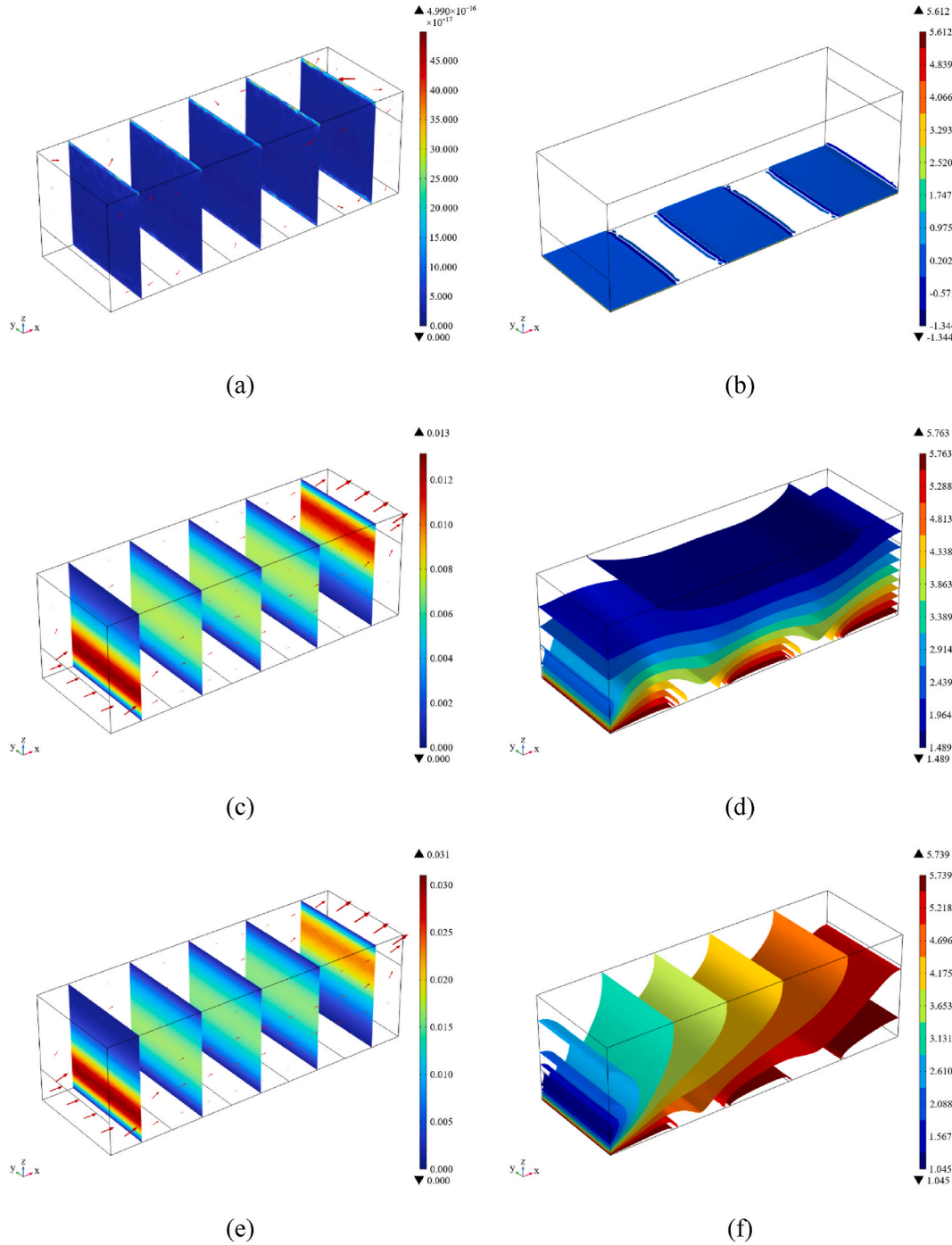
$$\rho \left( \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right), \quad (2)$$

$$\rho \left( \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left( \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right), \quad (3)$$

$$\rho \left( \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left( \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho g \beta (T - T_0), \quad (4)$$

$$\rho c_p \left( \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) = k \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right), \quad (5)$$

where, the independent variables  $x$ ,  $y$ ,  $z$ , and  $t$  denote the spatial (Cartesian) coordinates and time, respectively, and the dependent variables  $u$ ,  $v$ , and  $w$  denote the velocity components along the  $x$ -,  $y$ - and  $z$ -directions, respectively. The other dependent variables are temperature ( $T$ ), pressure ( $p$ ), and velocity components. The thermo-physical properties of air used in the equations are the density ( $\rho$ ), thermal conductivity ( $k$ ), specific heat at constant pressure ( $c_p$ ), dynamic viscosity ( $\mu$ ), and volumetric expansion coefficient ( $\beta$ ). These properties listed in Table 1 are assumed to be constant in the analysis. The thermal and flow fields are analyzed under well-imposed



**Fig. 6.** (a), (c), (e) Vertical slice of velocity magnitude plots and (b), (d), (f) isothermal surface plots in response to a P controller ( $K_p = 0.001 \text{ ms}^{-1} \text{K}^{-1}$ ) for (a), (b)  $t = 0$  s, (c), (d)  $t = 1$  s, and (e), (f)  $t = 5$  s (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

initial and boundary conditions listed in Table 2. Here,  $n$  is used as a generalized direction normal to the wall.

The controller action of the system is initiated depending on the value of  $T_p$ , and  $u_i$  reaches a non-zero value provided that  $T_p > T_{set}$ , which eventually causes a change in  $T_p$ , and the loop continues. The feedback system runs until  $T_p$  becomes less than or equal to  $T_{set}$ . The temporal variation of  $T_p$  is regarded as the response of the controller, which can be normalized as a non-dimensional temperature  $\theta$  by the following expression:

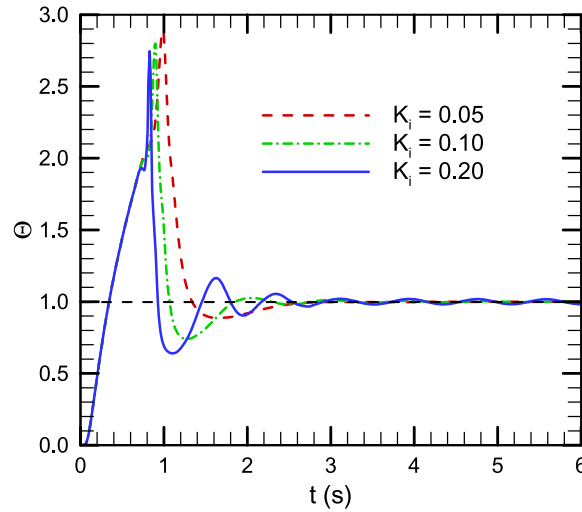


Fig. 7. System response in terms of  $\Theta$  at the probe point with time due to PI controller action for different  $K_i$  values at fixed  $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$  (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

$$\Theta = \frac{T_p - T_0}{T_{set} - T_0}. \quad (6)$$

#### 4. Numerical simulation and validation

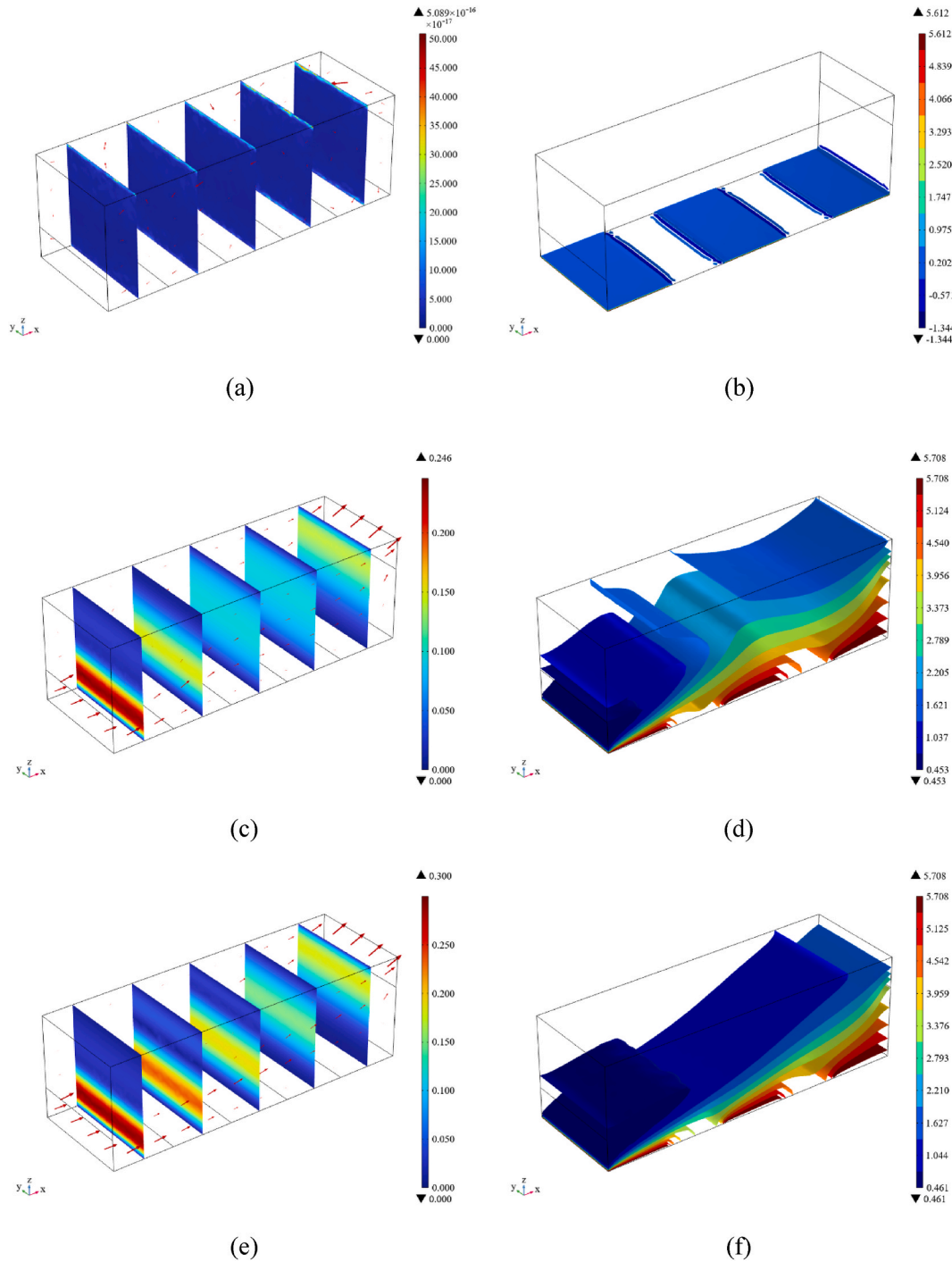
The governing equations (1)–(5), representing the nonlinear system of partial differential equations along with well-imposed initial and boundary conditions, are solved using commercial software “COMSOL Multiphysics 6.1”. The software employs a weighted residual method called the Galerkin finite element method to handle the nonlinearities associated with those systems of equations. This method is widely used in modern computational studies to solve the governing nonlinear differential equations [44,45]. Its discretization principle divides the entire domain into a finite number of tetrahedral (3D) mesh elements. Then, the weighted residuals for each element are expressed in terms of integral forms rather than any differential term, and each of these terms is solved using the Gauss quadrature integration technique. The boundary conditions are then applied to the already deduced nonlinear algebraic equations, usually adapted into matrices form. In the final stage, those nonlinear equations (matrices) are iterated to get the values of the dependent variables implementing the Newton-Raphson method. For this particular transient problem, a time-dependent solver is configured for time stepping (generalized alpha) and tolerance limit. A mesh independence test is done by taking  $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$ ,  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$ , and  $K_d = 0.01 \text{ mK}^{-1}$  and evaluating  $\Theta$  corresponding to a particular time  $t = 1.5 \text{ s}$  for varying the mesh configuration. Acquired  $\Theta$  values are then plotted against the number of elements ( $N_E$ ) for each mesh configuration shown in Fig. 3. The optimum mesh element is chosen at  $N_E = 373,970$ , for which the dependent variable ( $\Theta$ ) becomes almost constant. Also, the present numerical scheme is validated against a recent work done by Shuvo et al. [33] considering the variation of  $\Theta$  with respect to time ( $t$ ), at  $K_p = 0.01 \text{ ms}^{-1}\text{K}^{-1}$ ,  $K_i = 0.1 \text{ ms}^{-2}\text{K}^{-1}$ , and  $K_d = 0.001 \text{ mK}^{-1}$ . In that relative comparison, the present model is in good agreement, as shown in Fig. 4.

#### 5. Results and discussions

This study regulates the temperature inside the rectangular cuboid domain by setting a fixed temperature ( $T_{set}$ ) at a particular point ( $0.5L, 0.5W, 0.5H$ ). A continuous time-varying temperature ( $T_p$ ) at that point is monitored through a temperature probe (sensor), and the difference between these two temperatures ( $T_{set} - T_p$ ) is passed to the controller as an error signal ( $e$ ). Depending on the value of the error signals and constant gains ( $K_p$ ,  $K_i$ , and  $K_d$ ), P, PI, and PID controller actions are evaluated in terms of the output response ( $\Theta$ ) as a function of time. It is worth mentioning that I and D controllers will not be allowed to initiate a controlling action solely unless the P controller is present there. In the following subsections, the function of each of the proposed three controllers is discussed in more detail, along with qualitative visualizations in terms of 3D velocity magnitude and temperature surface plots representing the flow and thermal fields, respectively. Color legends correspond to the magnitudes of the dependent variables for both visualizations, while pointing arrows between planes define the direction of the flow.

##### 5.1. Action of P controller

Since the P controller only has non-zero proportional gain, it provides an output response proportional to the error signal, i.e., the difference between the output value where it should be and where it is. The proportional controller is the simplest form of a continuous feedback control system that offers a quick response or even sometimes an oscillation-free response. However, the controller's limitation is that it never brings the system to the desired set point, i.e., it has a steady-state error.



**Fig. 8.** (a), (c), (e) Vertical slice of velocity magnitude plots and (b), (d), (f) isothermal surface plots in response to a PI controller ( $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$  and  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$ ) for (a), (b)  $t = 0 \text{ s}$ , (c), (d)  $t = 1 \text{ s}$ , and (e), (f)  $t = 5 \text{ s}$  (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In Fig. 5, the response via the variation of  $\theta$  due to the proportional controller action is shown for the parametric changes in the proportional gain ( $K_p = 0.001, 0.005$ , and  $0.01 \text{ ms}^{-1}\text{K}^{-1}$ ). It is found that a lower value of  $K_p$  is associated with a higher steady-state error with lower oscillation and vice-versa. It also signifies the trade-off between accuracy and stability. When the accuracy becomes more elevated, there is a reduction in system stability. Therefore, at a lower value of  $K_p$  ( $=0.001 \text{ ms}^{-1}\text{K}^{-1}$ ), the steady-state error becomes almost 435% without oscillation. On the other hand, for  $K_p = 0.005\text{--}0.01 \text{ ms}^{-1}\text{K}^{-1}$ , the steady-state error reduces from 270% to 215% with an increase in oscillation. Similarly, settling time reduces from 2.6 to 1.5 s when a change in  $K_p$  occurs from 0.001 to 0.01

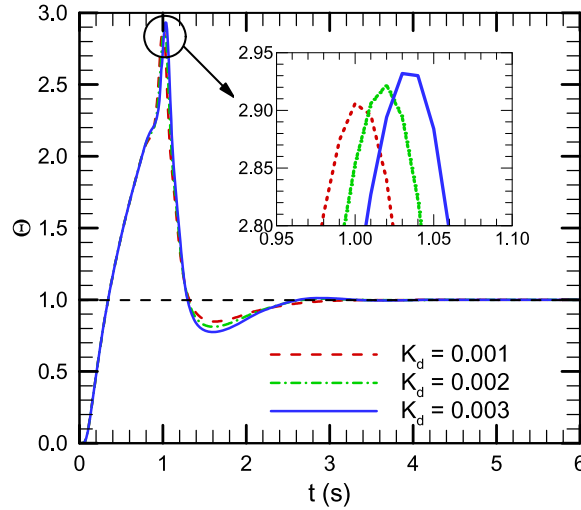


Fig. 9. System response in terms of  $\theta$  at the probe point with time due to PID controller action for different  $K_d$  values at fixed  $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$  and  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$  (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

$\text{ms}^{-1}\text{K}^{-1}$ . It concludes that a higher value of  $K_p$  is responsible for quick settling.

Moreover, P controller actions are qualitatively visualized in Fig. 6 in terms of the velocity magnitude and isothermal surface plots for  $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$  at three different time instances ( $t = 0, 1$ , and  $5 \text{ s}$ ). From Figs. 6 (a), (c), and (e), it is found that the flow velocity increases from almost zero to a maximum magnitude of  $0.031 \text{ ms}^{-1}$ . Besides, when controller action is initiated, the initial arbitrary directional flow turns out to be a regular flow toward the direction from the inlet to the outlet. The isothermal surface plot, on the other hand, shown in Figs. 6 (b), (d), and (f), dictates the gradual rise in domain temperature with time with no such tendency of the controller to settle it down near the desired condition. Therefore, it is observed that the final temperature value deviates significantly from the desired temperature.

### 5.2. Action of PI controller

The PI controller has a non-zero proportional and integral gain, and it employs a summation of both proportional and integral action of the error signal to have the final output response. The controller's integral action helps to vanish the steady-state error even though it takes longer to settle down. Therefore, combining I with P suffices to achieve a better-performing controller design.

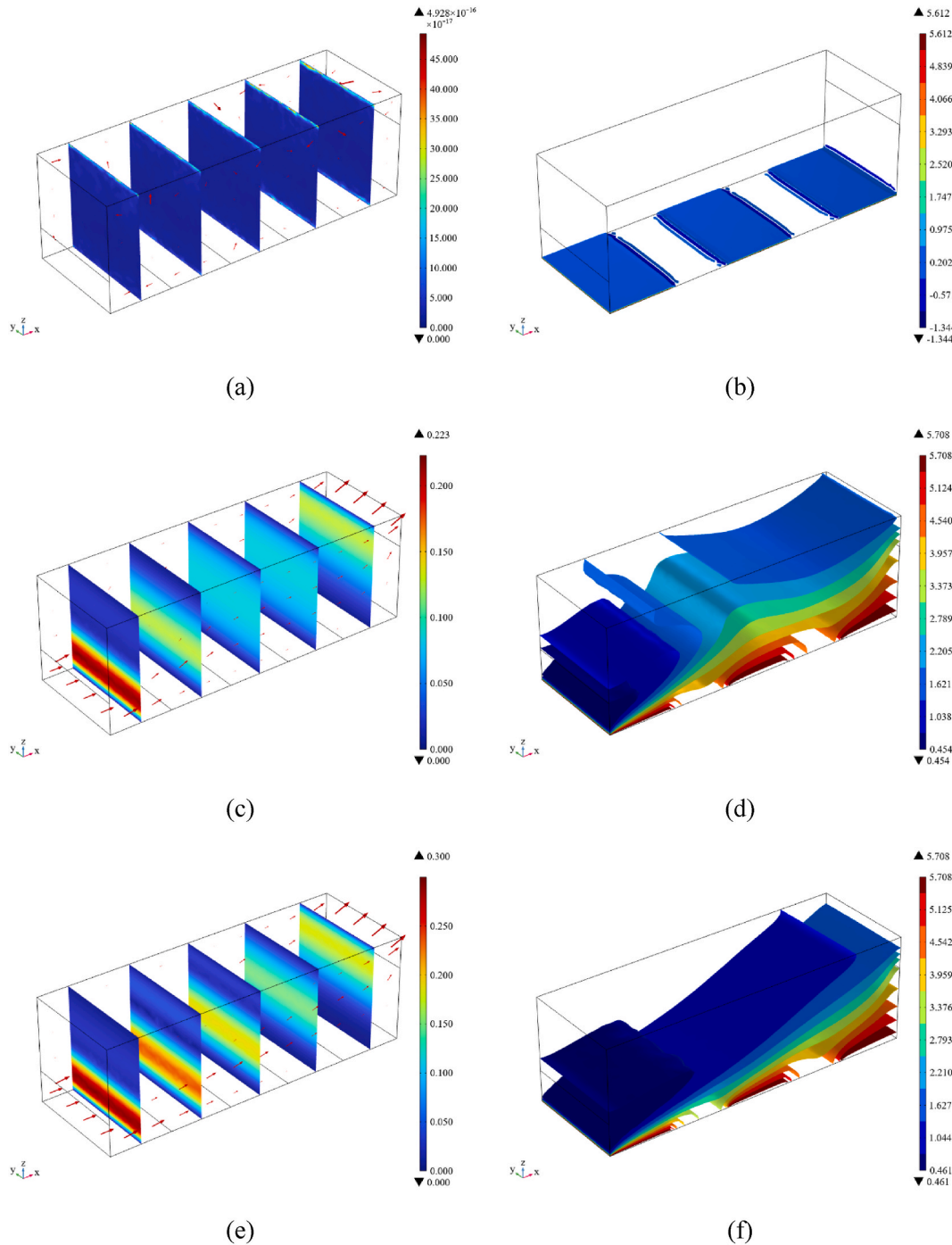
In this section, taking the most offset response from the previous P controller and keeping the setting to a fixed value of  $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$ , changes are only made for different integral gains ( $K_i = 0.05, 0.1$ , and  $0.2 \text{ ms}^{-2}\text{K}^{-1}$ ) to observe the time-varying output response shown in Fig. 7. From the response, it is found that higher  $K_i$  values are associated with higher oscillation and thus longer settling time. On the contrary, though lower  $K_i$  offers less oscillation and takes less time to settle down, it has maximum overshooting. From  $K_i = 0.05\text{--}0.2 \text{ ms}^{-2}\text{K}^{-1}$ , oscillation increases, and corresponding settling time goes from 3 to 6 s while overshooting decreases from 280% to 250%, respectively.

PI controller actions are also qualitatively visualized in terms of the vertical slice of velocity magnitude and isothermal surface plots (see Fig. 8) for three different time instances ( $t = 0, 1$ , and  $5 \text{ s}$ ). The results are included only for fixed proportional and integral gains ( $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$  and  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$ ). Like the P controller, the initial velocity remains zero until the activation of three heating elements increases the domain temperature. When the domain temperature rises, controller action tries to regulate the inflow condition so that the temperature reaches the desired value. Hence, it is observed that at the settling temperature, the velocity magnitude undergoes a substantial increase, getting a maximum value of  $0.3 \text{ ms}^{-1}$ . This magnitude is considerably higher than the corresponding value obtained with the P controller, as illustrated in Figs. 8 (a), (c), and (e). The PI controller action is more prominent in the isothermal surface plot, as shown in Figs. 8 (b), (d), and (f). From  $t = 0\text{--}1 \text{ s}$ , the temperature inside the domain rises from zero to a value higher than the desired condition due to isothermal heating. Again, from  $t = 1\text{--}5 \text{ s}$ , the controller action is performed in a way that minimizes the rise and ultimately settles it down at the desired condition.

### 5.3. Action of PID controller

Although the steady-state error reduces, the oscillation and settling time increase from P to PI controller. A PID controller is, therefore, introduced to have more optimum performance and a better version of all controllers. In this context, the D represents a derivative action, which differs from both P and I controllers as it functions as a feed-forward controller. Additionally, all the gains associated with a PID controller occupy non-zero values. It operates mainly to diminish the change in error by keeping the system in the desired setting. Alongside, under appropriate tuning, it effectively reduces overshoot and oscillation.

In this particular section, the effects of different derivative gains ( $K_d = 0.001, 0.002$ , and  $0.003 \text{ ms}^{-1}\text{K}^{-1}$ ) are analyzed by taking the maximum overshoot response setting from the PI controller (i.e.,  $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$  and  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$ ) as shown in Fig. 9. It can



**Fig. 10.** (a), (c), (e) Vertical slice of velocity magnitude plots and (b), (d), (f) isothermal surface plots in response to a PID controller ( $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$ ,  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$ , and  $K_d = 0.001 \text{ mK}^{-1}$ ) for (a), (b)  $t = 0$  s, (c), (d)  $t = 1$  s, and (e), (f)  $t = 5$  s (color online). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

be seen that the overshoot decreases a little with the improvement in oscillation and settling time. Additionally, the change in  $K_d$  from 0.001 to 0.003  $\text{mK}^{-1}$  shows a negligible disparity. However, if the optimum one is to choose, PID with  $K_d = 0.001 \text{ mK}^{-1}$  would suffice. Upon closer examination of the maximum overshooting data for various differential gains, it becomes evident that tuning offers significant advantages since a decreasing pattern of overshooting is observed.

The qualitative visualizations for PID controller action are shown in Fig. 10, similar to P and PI controllers adopting a fixed optimum setting ( $K_p = 0.001 \text{ ms}^{-1}\text{K}^{-1}$ ,  $K_i = 0.05 \text{ ms}^{-2}\text{K}^{-1}$ , and  $K_d = 0.001 \text{ mK}^{-1}$ ). Since the visualizations for three different time

instances are displayed, the underlying controlling action that results in an oscillatory response remains unnoticed. Apart from that, PID controller action reacts similarly to the PI controller regarding flow or thermal field visualization.

## 6. Conclusion

This study investigates thermal management inside a discrete, isothermally heated rectangular cuboid through the active control actions of P, PI, and PID closed-loop controllers. The following statements are deemed crucial obtained from this analysis:

- i. Solely employing a P controller action fails to eliminate steady-state error, consequently hindering the attainment of a regulated temperature condition within the system domain. Moreover, while a higher gain of the P controller enhances accuracy, it compromises stability to a significant extent.
- ii. By implementing a PI controller action, zero steady-state error can be achieved while ensuring faster settling. Lower integral gains result in a quicker response and reduced oscillation, albeit with a slightly higher risk of overshooting.
- iii. Implementing a PID controller action exhibits similarities to the PI controller action. However, lower derivative gains have been observed to enhance oscillation control and reduce response time. An optimal system response can be achieved by appropriately tuning the gains, facilitating improved thermal system management.
- iv. When the controller actions are initiated, the initial arbitrary directional flow transforms into a regular flow from the inlet to the outlet. The isothermal surface plot shows a gradual upward shift over time, with the P controller failing to stabilize it near the desired condition. In contrast, both PI and PID controllers act to minimize this upward shift in the isothermal surface within the domain and eventually settle it at the desired condition.

The study's novelty stems from its potential as a benchmark model for designing controlled environments and thermal management in various applications [46], such as MEMS fabrication, electronic component cooling, drug manufacturing facilities, etc. By utilizing the insights gained from this study, researchers and engineers can enhance their ability to create optimal and regulated conditions in these domains, leading to improved efficiency and reliability.

## CRediT authorship contribution statement

**Niloy Deb:** Data curation, Modeling, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Sadib Fardin:** Conceptualization, Literature review, Formal analysis, Software, Visualization, Writing – review & editing. **Md Muhtasim Fardin:** Data curation, Modeling, Methodology, Formal analysis, Software, Visualization. **Nudrat Nawal:** Writing – review & editing. **Md Rashed Nizam:** Writing – review & editing. **Sumon Saha:** Supervision, Formal analysis, Software, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Nomenclature

$c_p$ : Specific heat capacity ( $\text{J kg}^{-1}\text{K}^{-1}$ )  
 $e$ : Error signal (K)  
 $g$ : Gravitation acceleration ( $\text{ms}^{-2}$ )  
 $H$ : Height of cuboid (m)  
 $k$ : Thermal conductivity ( $\text{W m}^{-1}\text{K}^{-1}$ )  
 $K_d$ : Derivative gain ( $\text{mK}^{-1}$ )  
 $K_i$ : Integral gain ( $\text{ms}^{-2}\text{K}^{-1}$ )  
 $K_p$ : Proportional gain ( $\text{ms}^{-1}\text{K}^{-1}$ )  
 $L$ : Length of cuboid (m)  
 $n$ : Generalized wall normal direction (m)  
 $N_E$ : Number of elements  
 $p$ : Pressure (Pa)  
 $T$ : Temperature (K)  
 $t$ : Time (s)  
 $T_0$ : Initial domain temperature (K)  
 $T_h$ : Heated wall temperature (K)  
 $T_i$ : Inlet temperature (K)  
 $T_p$ : Probe-point temperature (K)  
 $T_{set}$ : Set point temperature (K)

$u, v, w$ : Velocity components ( $\text{ms}^{-1}$ )  
 $u_i$ : Inlet air velocity ( $\text{ms}^{-1}$ )  
 $W$ : Width of cuboid (m)  
 $x, y, z$ : Cartesian coordinates (m)

### *Greek symbols*

$\beta$ : Coefficient of volumetric expansion ( $\text{K}^{-1}$ )  
 $\theta$ : Dimensionless temperature  
 $\lambda$ : Geometric distance parameter  
 $\mu$ : Dynamic viscosity (Pa.s)  
 $\rho$ : Density ( $\text{kgm}^{-3}$ )

### *Abbreviations*

$D$ : Derivative  
 $I$ : Integral  
 $P$ : Proportional